

Lab Assignment 2

Object-Oriented Programming (CSD213/2004)

Topic: Arithmetic Expressions, Conditional Statements

Total Marks: For practice only

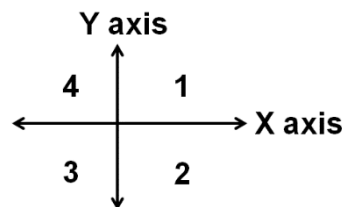
Deadline: Before next Lab (P1 batch: 31-08-2026 | 1:00 pm)

Submission Details: Write all programs using an IDE (e.g., NetBeans). Each program file must be named in the format <ProgramNumber>_<StudentRollNo>.java. For example, if your roll number is 241000189, the filenames should be:

- Program1_241000189.java
- Program2_241000189.java
- Program3_241000189.java, and so on.

After completing all programs, compress (zip) all the files into a single archive named **ProgAssn2-<StudentRollNo>.zip**. Finally, upload the zip file to the Blackboard page for this course.

1. Read a pair of integers representing the x and y coordinates of a point **p** and display a message telling if the point **p** lies either on the x-axis or the y-axis (or both). If it neither lies on x axis nor on y axis, then display a message telling in which quadrant it lies (the quadrants are numbered as shown below).



2. Write a program to read three integer values from the user. Display the three values, determine the maximum, and display it.
3. Read the total marks (Integer) of a student in a subject out of 100. Display his/her grade in the subject. Use the following Marks to Grade conversion table:

Mark>=90	Grade=EX
Mark>=80	Grade=A
Mark>=70	Grade=B
Mark>=60	Grade=C

Mark>=50 Grade=D
Mark>=35 Grade=P,
Mark<35 Grade=F

4. Read four floating-point numbers **m1,c1,m2,c2** that represent two straight lines of the form **y=mx+c**. Check if the two lines intersect or if they are parallel. Display your result. If they intersect, display their point of intersection.
5. Assume that a class has 5 students. Read the Roll number (integer) and Marks out of (100) for the 5 students. Compute the average mark (float) for each student and display the data nicely formatted.

Example:

Enter details for student 1:

Roll: 1001

Marks: 75 35 95 50 99

Enter details for student 2:

Roll: 2435

Marks: 50 45 99 80 83

Enter details for student 3:

Roll: 1023

Marks: 75 55 89 77 56

Enter details for student 4:

Roll: 2132

Marks: 88 77 98 78 64

Enter details for student 5:

Roll: 3152

Marks: 54 20 21 86 55

Tabular Display:

Roll:	1001	Marks: 75	35	95	50	99	Average= 70.8
Roll:	2435	Marks: 50	45	99	80	83	Average= 71.4
Roll:	1023	Marks: 75	55	89	77	56	Average= 70.4
Roll:	2132	Marks: 88	77	98	78	64	Average= 81
Roll:	3152	Marks: 54	20	21	86	55	Average= 47.2

Note: Use of Loops and Arrays in your program will carry 50% penalty